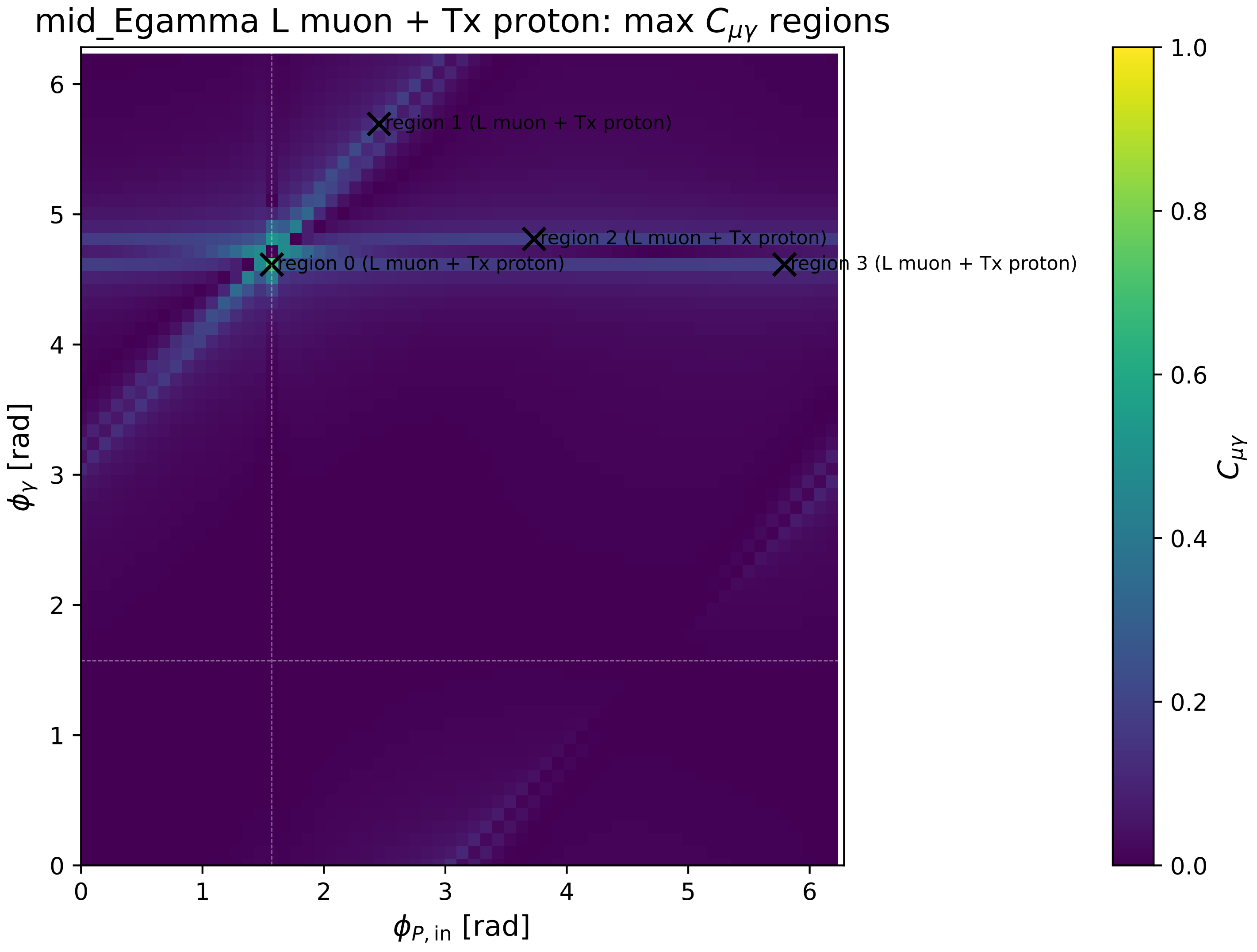
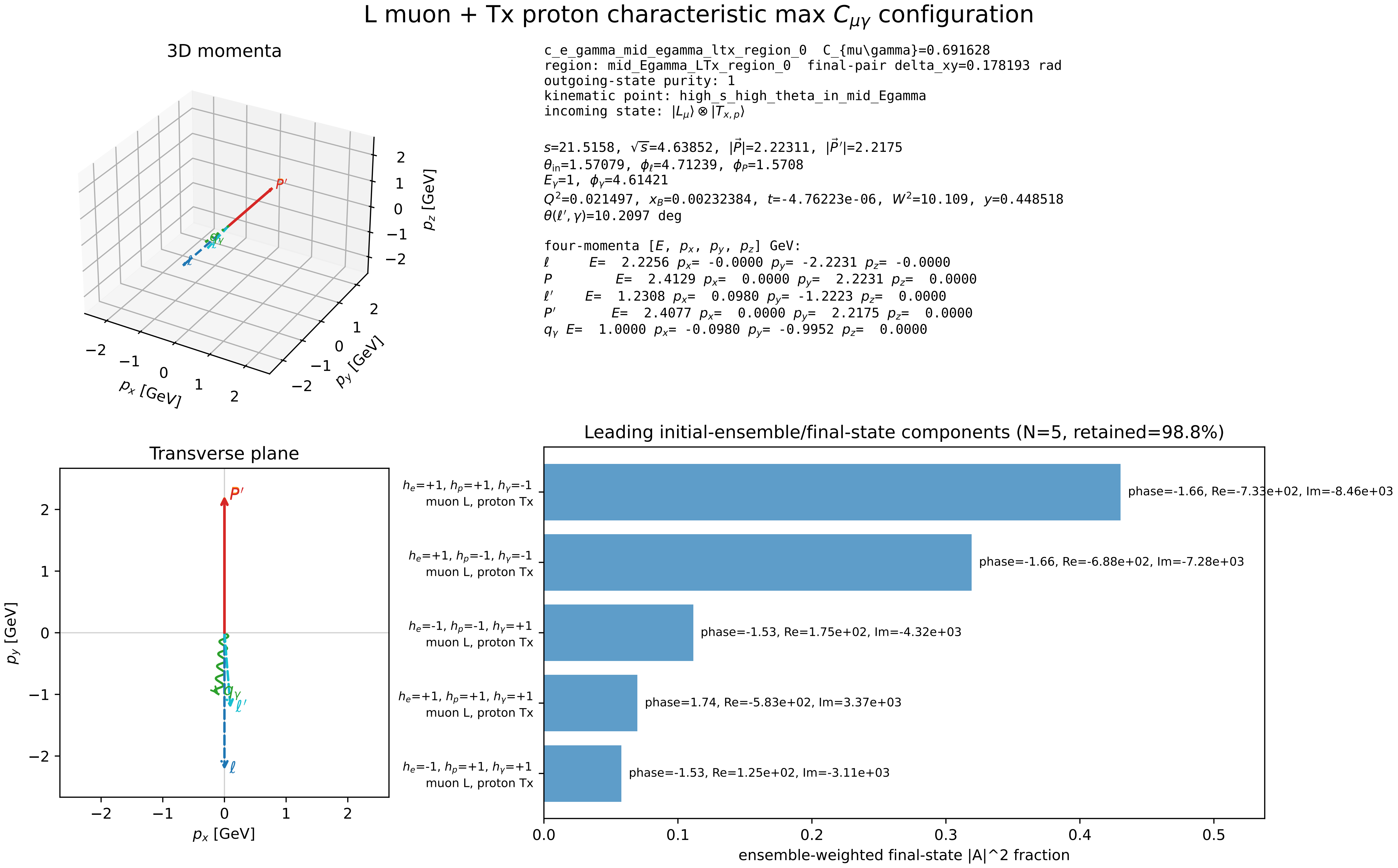
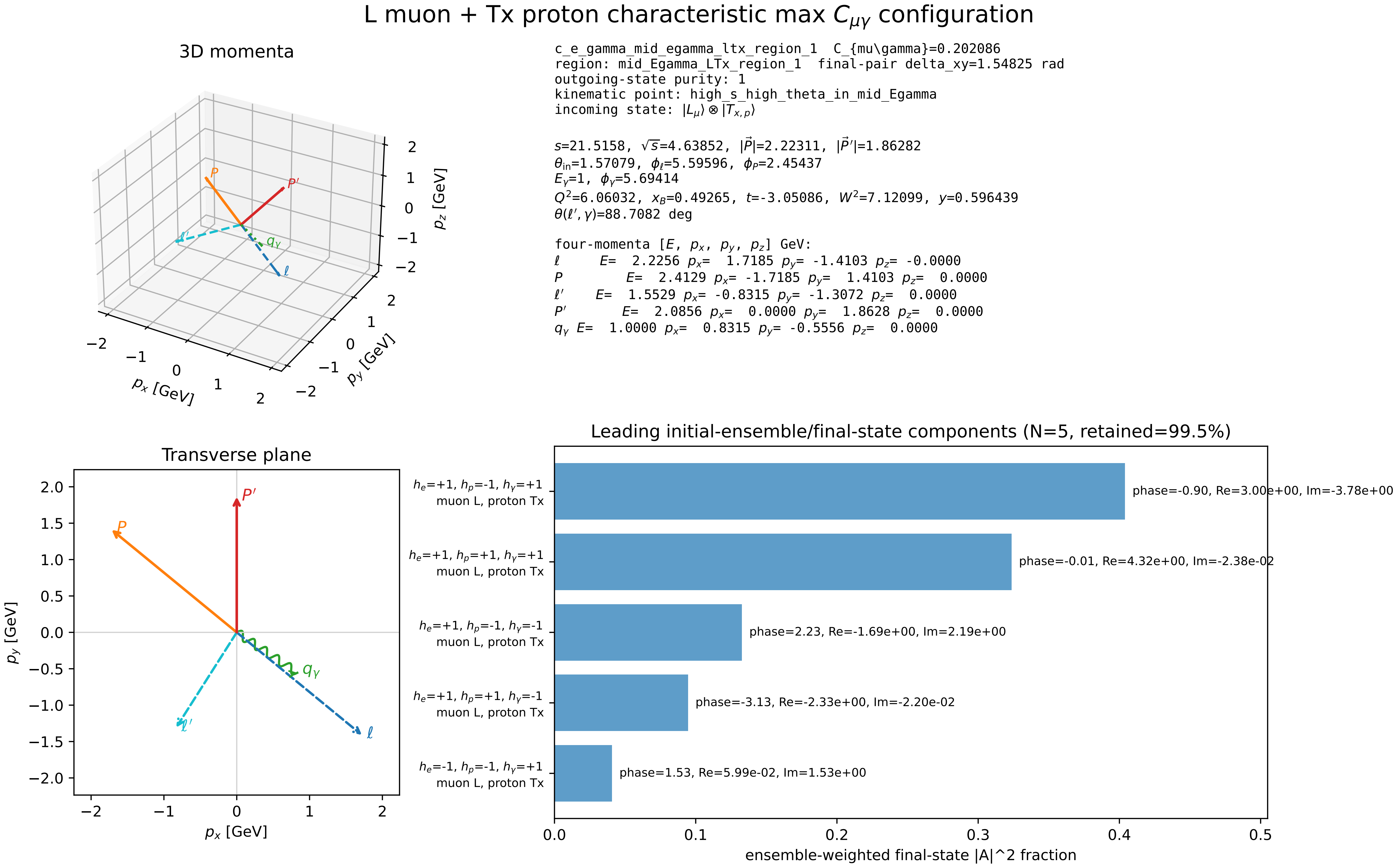


mid_Egamma L muon + Tx proton: max $C_{\mu\gamma}$ regions
region 1 (L muon + Tx proton)
region 2 (L muon + Tx proton)
region 0 (L muon + Tx proton)
region 3 (L muon + Tx proton)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
$C_{\mu\gamma}$
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

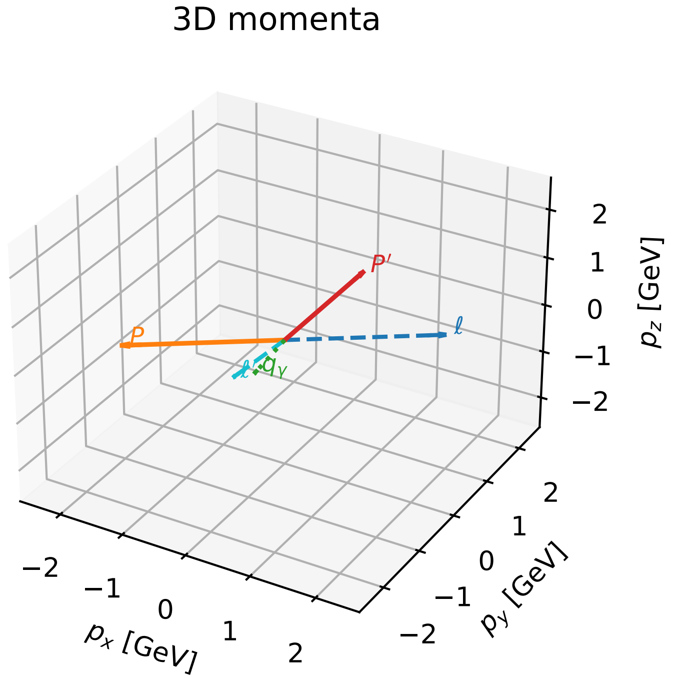


L muon + Tx proton characteristic max $C_{\mu\gamma}$ configuration
3D momenta
c_e_gamma_mid_egamma_ltx_region_0 C_{mu\gamma}=0.691628
region: mid_Egamma_LTx_region_0 final-pair delta_xy=0.178193 rad
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $|L_\mu\rangle \otimes |T_{x,p}\rangle$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=2.2175
$\theta_{\rm in}$=1.57079, ϕ_ℓ=4.71239, ϕ_P=1.5708
E_γ=1, ϕ_γ=4.61421
Q^2=0.021497, x_B=0.00232384, t=-4.76223e-06, W^2=10.109, y=0.448518
$\theta(\ell',\gamma)$=10.2097 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= -0.0000 p_y= -2.2231 p_z= -0.0000
P E= 2.4129 p_x= 0.0000 p_y= 2.2231 p_z= 0.0000
ℓ' E= 1.2308 p_x= 0.0980 p_y= -1.2223 p_z= 0.0000
P' E= 2.4077 p_x= 0.0000 p_y= 2.2175 p_z= 0.0000
q_γ E= 1.0000 p_x= -0.0980 p_y= -0.9952 p_z= 0.0000
p_x [GeV]
p_y [GeV]
p_z [GeV]
P'
q_γ
ℓ'
ℓ
Transverse plane
Leading initial-ensemble/final-state components (N=5, retained=98.8%)
h_e=+1, h_p=+1, h_γ=-1 muon L, proton Tx
phase=-1.66, Re=-7.33e+02, Im=-8.46e+03
h_e=+1, h_p=-1, h_γ=-1 muon L, proton Tx
phase=-1.66, Re=-6.88e+02, Im=-7.28e+03
h_e=-1, h_p=-1, h_γ=+1 muon L, proton Tx
phase=-1.53, Re=1.75e+02, Im=-4.32e+03
h_e=+1, h_p=+1, h_γ=+1 muon L, proton Tx
phase=1.74, Re=-5.83e+02, Im=3.37e+03
h_e=-1, h_p=+1, h_γ=+1 muon L, proton Tx
phase=-1.53, Re=1.25e+02, Im=-3.11e+03
ensemble-weighted final-state |A|^2 fraction



L muon + Tx proton characteristic max $C_{\mu\gamma}$ configuration
3D momenta
c_e_gamma_mid_egamma_ltx_region_1 C_{mu\gamma}=0.202086
region: mid_Egamma_LTx_region_1 final-pair delta_xy=1.54825 rad
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $|L_\mu\rangle \otimes |T_{x,p}\rangle$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=1.86282
θ_{in}=1.57079, ϕ_ℓ=5.59596, ϕ_P=2.45437
E_γ=1, ϕ_γ=5.69414
Q^2=6.06032, x_B=0.49265, t=-3.05086, W^2=7.12099, y=0.596439
$\theta(\ell',\gamma)$=88.7082 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= 1.7185 p_y= -1.4103 p_z= -0.0000
P E= 2.4129 p_x= -1.7185 p_y= 1.4103 p_z= 0.0000
ℓ' E= 1.5529 p_x= -0.8315 p_y= -1.3072 p_z= 0.0000
P' E= 2.0856 p_x= 0.0000 p_y= 1.8628 p_z= 0.0000
q_γ E= 1.0000 p_x= 0.8315 p_y= -0.5556 p_z= 0.0000
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
Leading initial-ensemble/final-state components (N=5, retained=99.5%)
h_e=+1, h_p=-1, h_γ=+1 muon L, proton Tx
phase=-0.90, Re=3.00e+00, Im=-3.78e+00
h_e=+1, h_p=+1, h_γ=+1 muon L, proton Tx
phase=-0.01, Re=4.32e+00, Im=-2.38e-02
h_e=+1, h_p=-1, h_γ=-1 muon L, proton Tx
phase=2.23, Re=-1.69e+00, Im=2.19e+00
h_e=+1, h_p=+1, h_γ=-1 muon L, proton Tx
phase=-3.13, Re=-2.33e+00, Im=-2.20e-02
h_e=-1, h_p=-1, h_γ=+1 muon L, proton Tx
phase=1.53, Re=5.99e-02, Im=1.53e+00
ensemble-weighted final-state |A|^2 fraction

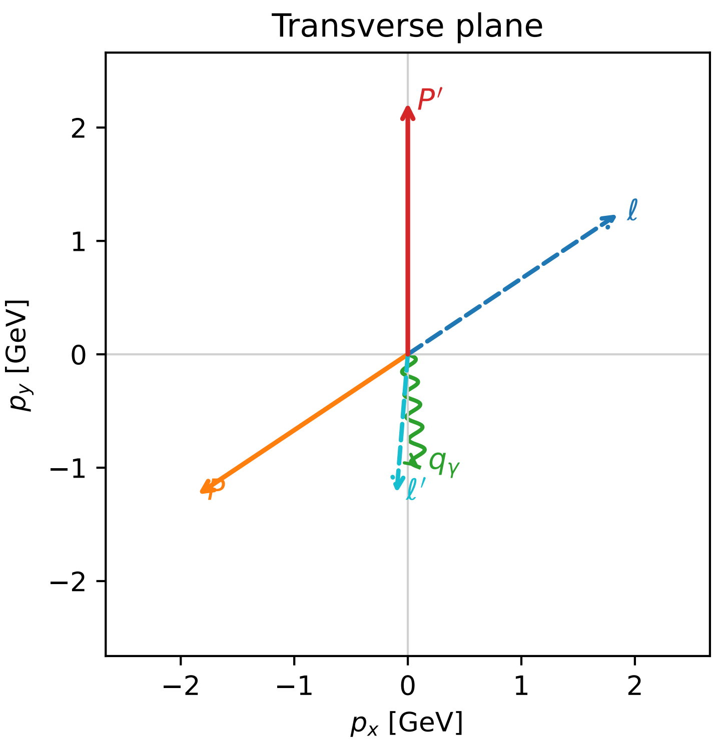
L muon + Tx proton characteristic max $C_{\mu\gamma}$ configuration

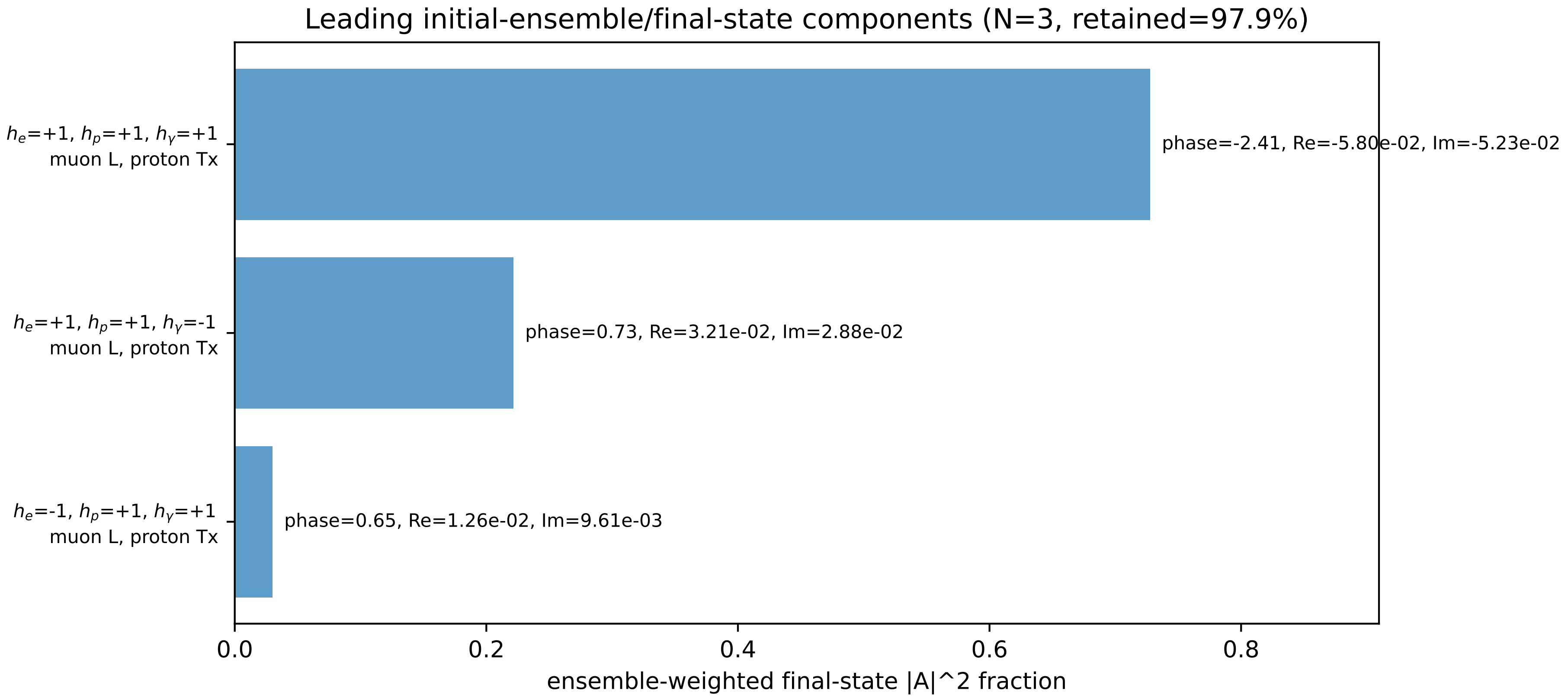


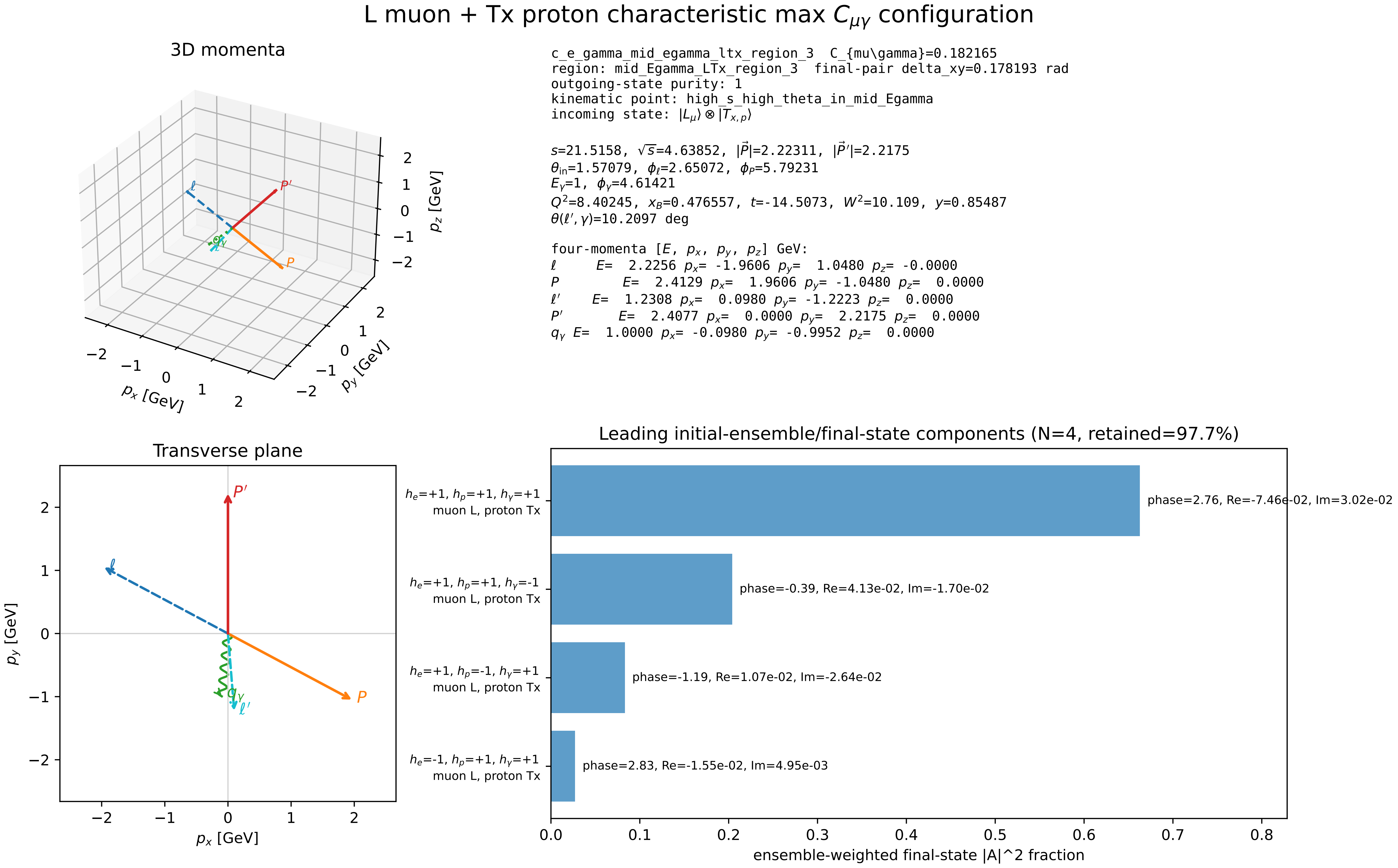
c_e_gamma_mid_egamma_ltx_region_2 C_{mu\gamma}=0.182232
region: mid_Egamma_LTx_region_2 final-pair delta_xy=0.178193 rad
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $|L_\mu\rangle \otimes |T_{x,p}\rangle$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=2.2175
θ_{in}=1.57079, ϕ_ℓ=0.589049, ϕ_P=3.73064
E_γ=1, ϕ_γ=4.81056
Q^2=8.83792, x_B=0.489174, t=-15.3372, W^2=10.109, y=0.875984
$\theta(\ell', \gamma)$=10.2097 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= 1.8484 p_y= 1.2351 p_z= -0.0000
P E= 2.4129 p_x= -1.8484 p_y= -1.2351 p_z= 0.0000
ℓ' E= 1.2308 p_x= -0.0980 p_y= -1.2223 p_z= 0.0000
P' E= 2.4077 p_x= 0.0000 p_y= 2.2175 p_z= 0.0000
q_γ E= 1.0000 p_x= 0.0980 p_y= -0.9952 p_z= 0.0000






L muon + Tx proton characteristic max $C_{\mu\gamma}$ configuration
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
ℓ
P'
P
q_γ
ℓ'
c_e_gamma_mid_egamma_ltx_region_3 C_{mu\gamma}=0.182165
region: mid_Egamma_LTx_region_3 final-pair delta_xy=0.178193 rad
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $|L_\mu\rangle \otimes |T_{x,p}\rangle$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=2.2175
θ_{in}=1.57079, ϕ_ℓ=2.65072, ϕ_P=5.79231
E_γ=1, ϕ_γ=4.61421
Q^2=8.40245, x_B=0.476557, t=-14.5073, W^2=10.109, y=0.85487
$\theta(\ell', \gamma)$=10.2097 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= -1.9606 p_y= 1.0480 p_z= -0.0000
P E= 2.4129 p_x= 1.9606 p_y= -1.0480 p_z= 0.0000
ℓ' E= 1.2308 p_x= 0.0980 p_y= -1.2223 p_z= 0.0000
P' E= 2.4077 p_x= 0.0000 p_y= 2.2175 p_z= 0.0000
q_γ E= 1.0000 p_x= -0.0980 p_y= -0.9952 p_z= 0.0000
Transverse plane
p_x [GeV]
p_y [GeV]
Leading initial-ensemble/final-state components (N=4, retained=97.7%)
h_e=+1, h_p=+1, h_γ=+1 muon L, proton Tx
phase=2.76, Re=-7.46e-02, Im=3.02e-02
h_e=+1, h_p=+1, h_γ=-1 muon L, proton Tx
phase=-0.39, Re=4.13e-02, Im=-1.70e-02
h_e=+1, h_p=-1, h_γ=+1 muon L, proton Tx
phase=-1.19, Re=1.07e-02, Im=-2.64e-02
h_e=-1, h_p=+1, h_γ=+1 muon L, proton Tx
phase=2.83, Re=-1.55e-02, Im=4.95e-03
ensemble-weighted final-state |A|^2 fraction